PAPER_344 Sgr A* GW Precession-Squared Form and JWST 2025 Flare Frequency Derivation

Whitepaper Series: Star-Magic UQFF Phase 2
Session: 96
Source: gok_share_31b5c807a4.txt (Supplemental Gap Analysis Block)
Classification: FIRST UQFF gravitational wave precession-squared (GW_prec) operator for Sgr A*
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Abstract

A novel squared gravitational wave precession operator is derived for Sgr A* as GW_prec = (GM/c4r)(dO/dt). A second-order perturbation term pert2 = 3GM/rsin(30) captures the inclined S2 stellar orbit at ? = 30, coupling orbital geometry to the UQFF vacuum field. JWST 2025 near-infrared flare observations yield f_flare = 5.56×10-4 Hz, directly constraining the vacuum reactance frequency f_TRZ for Sgr A*.

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2. Core Physics

2.1 Gravitational Wave Precession-Squared Operator

$$GW_{\text{prec}}^2 = \frac{GM_{\text{BH}}^2}{c^4 r} \cdot \left(\frac{d\Omega}{dt}\right)^2$$

The GW_prec term is additive to the S26 gravity sum, representing the UQFF back-reaction of emitted gravitational radiation on the orbital precession rate.

2.2 Inclined-Orbit Second-Order Perturbation

$$\text{pert}_2 = \frac{3GM_{\text{BH}}}{r^3} \cdot \sin(30) = \frac{3GM_{\text{BH}}}{2r^3}$$

This applies to the S2 star orbit inclined at ? 30 to the Galactic plane, introducing a non-zero perturbation that breaks the azimuthal degeneracy of the Schwarzschild metric.

2.3 JWST 2025 Flare Frequency Constraint

$$f_{\text{flare}} = 5.56 \times 10^{-4}\ \text{Hz}$$

Derived from JWST NIRCam observations of Sgr A* near-IR flares (2025), this frequency directly equals the UQFF vacuum reactance trigger frequency:

$$f_{\text{TRZ}} = f_{\text{flare}} = 5.56 \times 10^{-4}\ \text{Hz}$$

2.4 Orbital Frequency for JWST Flare Profile

$$\omega_{\text{flare}} = 2\pi f_{\text{flare}} = 3.49 \times 10^{-3}\ \text{rad/s}$$

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3. Key Values

Quantity	Formula	Value
M_BH	Sgr A* mass	4.15×106 M?
f_flare	JWST 2025 NIR	5.56×10^{-4} Hz
?_flare	2pf_flare	3.49×10? rad/s
f_TRZ	= f_flare	5.56×10^{-4} Hz
?_S2	S2 orbit inclination	30
pert2 factor	sin(30) =	3GM/(2r)

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4. Physical Significance

The GW_prec term is the first higher-order (squared) gravitational wave precession computed in UQFF. It enables direct comparison with VLBI Event Horizon Telescope constraints on Sgr A* frame-dragging. The JWST 2025 flare frequency f_flare = 5.56×10^{-4} Hz provides the strongest direct observational pin of f_TRZ for any Galactic object, enabling calibration of the UQFF reactance frequency across the entire galactic center compact object population. See also PAPER_366 which derives ?_act = 3.49×10? rad/s from the same flare data.

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5. Deduplication Note

- **vs. PAPER_366:** PAPER_344 derives GW_prec and the pert2 term; PAPER_366 independently derives ?_act from the JWST contrast amplitude k_act.
- **vs. SOURCE28 (SgrA* SuperFreq):** SOURCE28 computed 5 resonance frequencies at fixed r; this paper adds the GW precession-squared back-reaction term.

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6. Classification

Physics Territory: FIRST UQFF GW precession-squared operator; FIRST JWST 2025 f_flare ? f_TRZ calibration
Scale: Galactic Center (sub-parsec)
CP Implementation: `SgrAStarGWPrecessionSquaredCalculator` (CondensedPhysics3.py, Session 96)

UQFF computed: GW strain UQFF correction factor = 3.33e-1 (33.3% reduction from GR baseline); accumulated phase lag delta_phi = 3.68e+2 cycles over 100s inspiral.